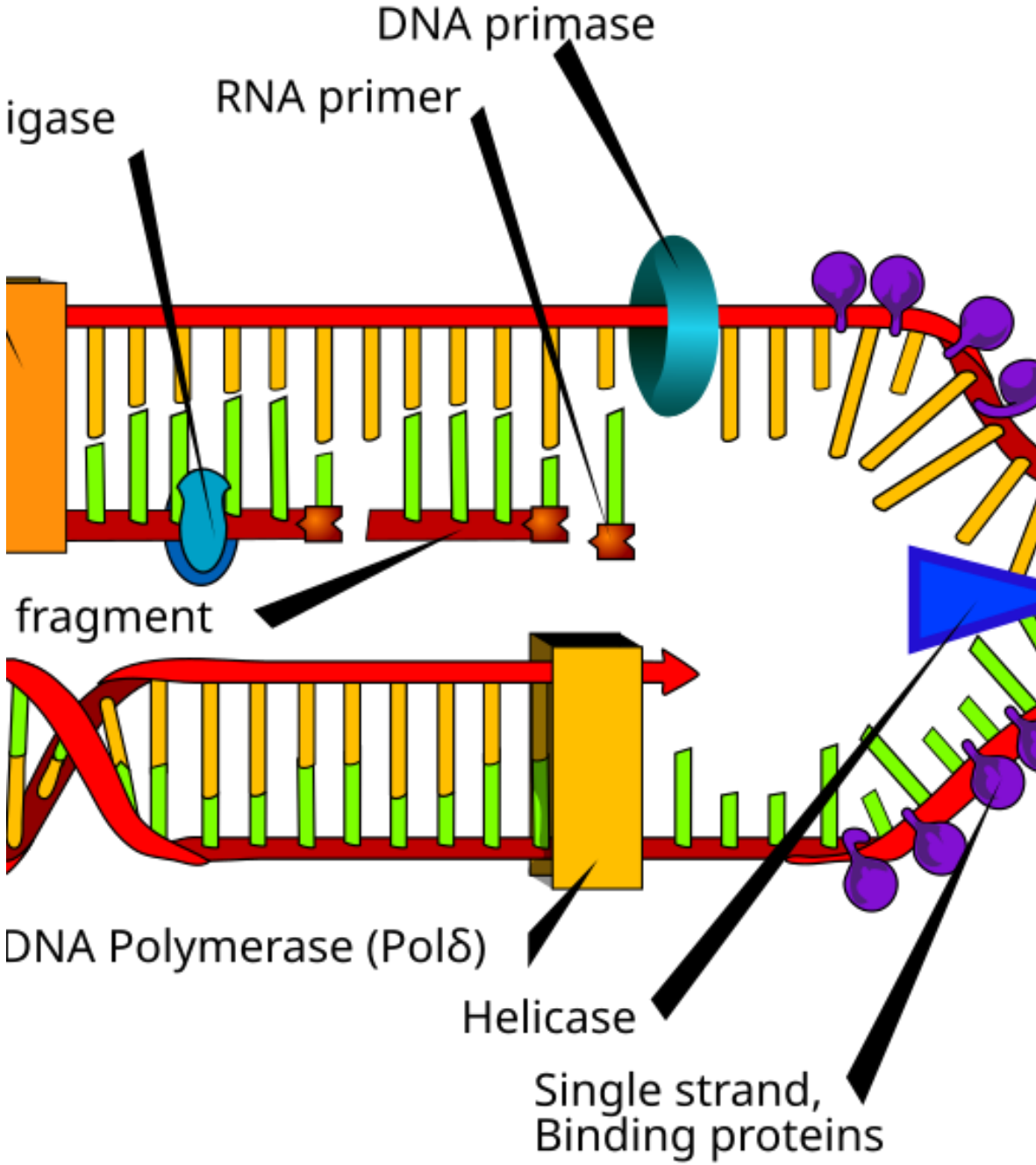
DP Biology – D1.1 (HL extension)

Inside DNA replication

- How do cells make new DNA?
- What biotechnology has understanding DNA replication made possible?



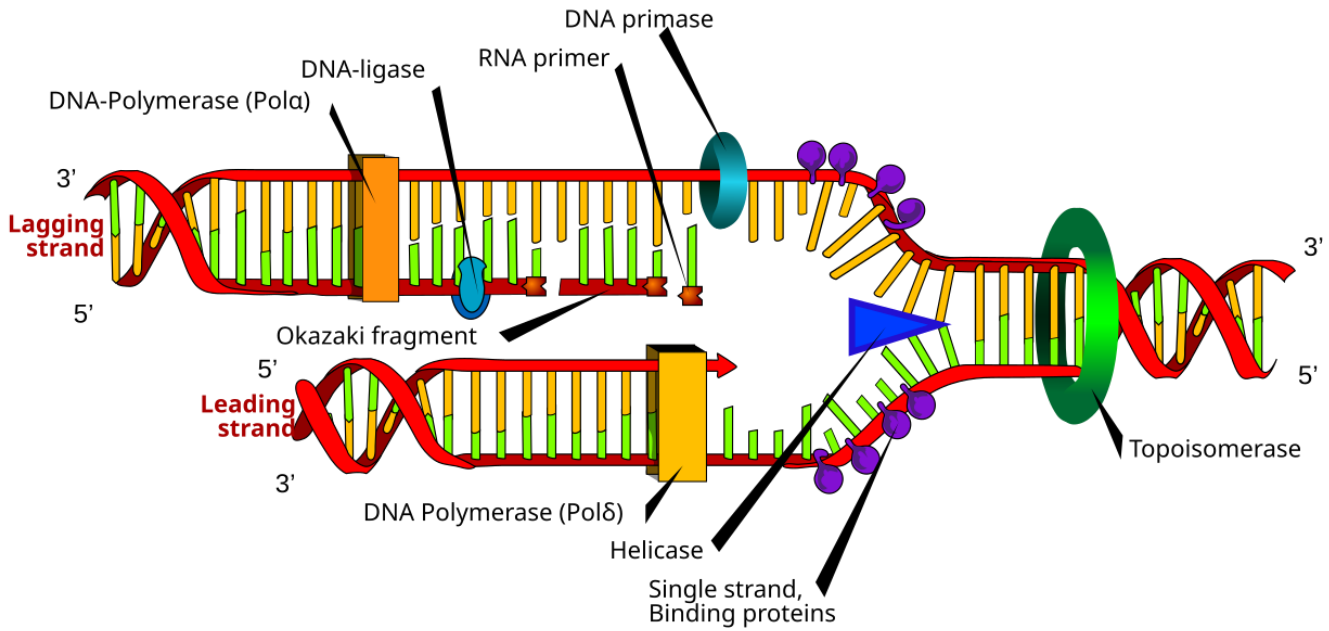
HL statements D1.1.6–9. Textbook pp. 604–614.
Image: DNA replication. Mariana Ruiz Villarreal, public domain.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	D1.1.6–7	Explain why DNA polymerases build 5′ to 3′, and compare the leading and lagging strands
2	D1.1.8	Explain the roles of helicase, primase, DNA polymerases I and III, and ligase
3	D1.1.9	Explain how DNA polymerase III proofreads, and why this matters

Recap: a replication fork

D1.1.6



A replication fork. Mariana Ruiz Villarreal, public domain.

At SL: helicase separates the strands; DNA polymerase builds new strands on each template.

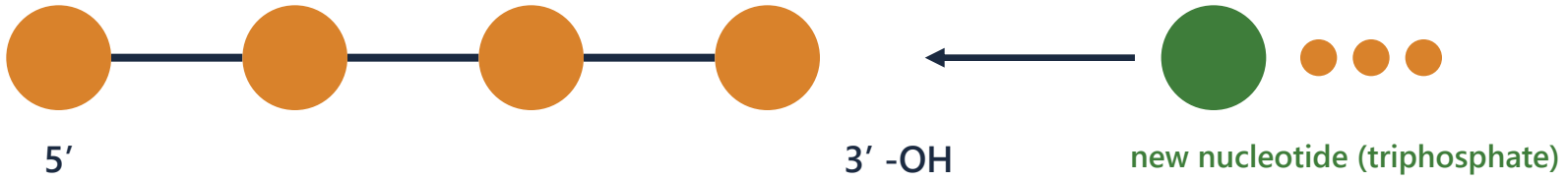
At HL: **why** the two new strands are made differently, and which enzymes each one needs.

Question for today

Why is one new strand made in pieces?

DNA polymerase works 5′ to 3′

D1.1.6

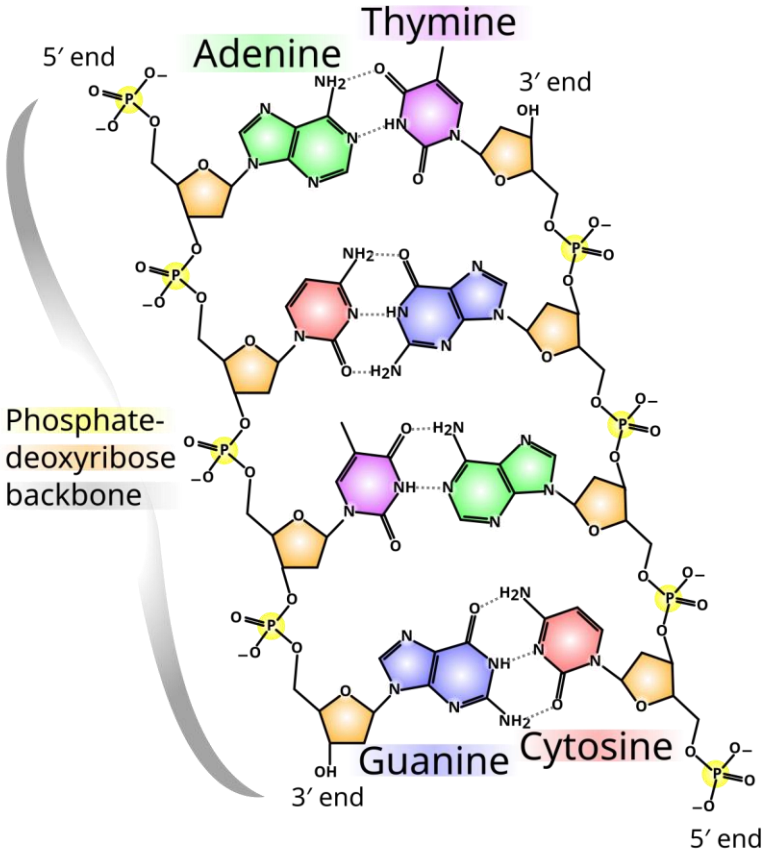


A new nucleotide arrives with **three phosphates**. Polymerase joins its 5′ phosphate to the free **3′ -OH** at the end of the strand; two phosphates are released, giving the energy.

So new DNA always grows **5′ to 3′**, and polymerase cannot start a strand from nothing.

Why direction causes a problem

D1.1.6



DNA strands are antiparallel. CC BY-SA 3.0.

The two strands of DNA run in **opposite directions** (antiparallel).

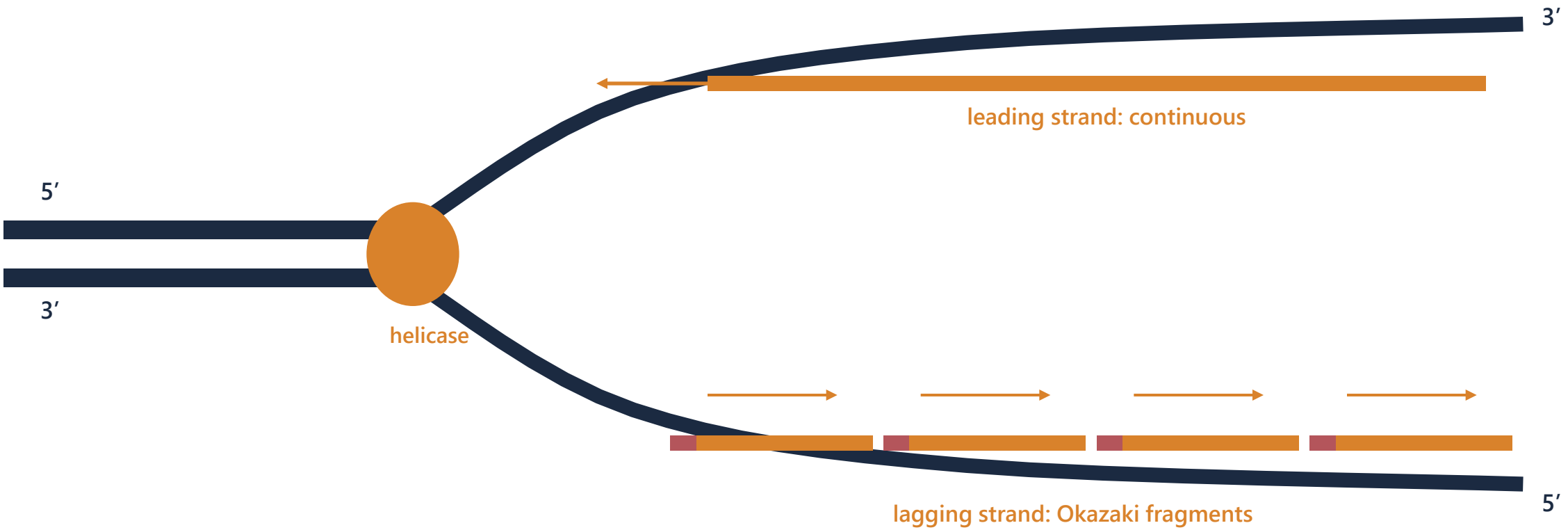
As the fork opens, one template lets a new strand grow towards the fork. The other template runs the wrong way, so its new strand must grow **away** from the fork.

Think

Why can’t polymerase simply build the second strand towards the fork too?

Leading and lagging strands

D1.1.7



Drawn fork. Navy: templates. Orange: new DNA. Red: RNA primers. The fork moves left.

Key idea

Leading strand: made continuously towards the fork. Lagging strand: made in short Okazaki fragments away from it.

Leading and lagging compared

D1.1.7

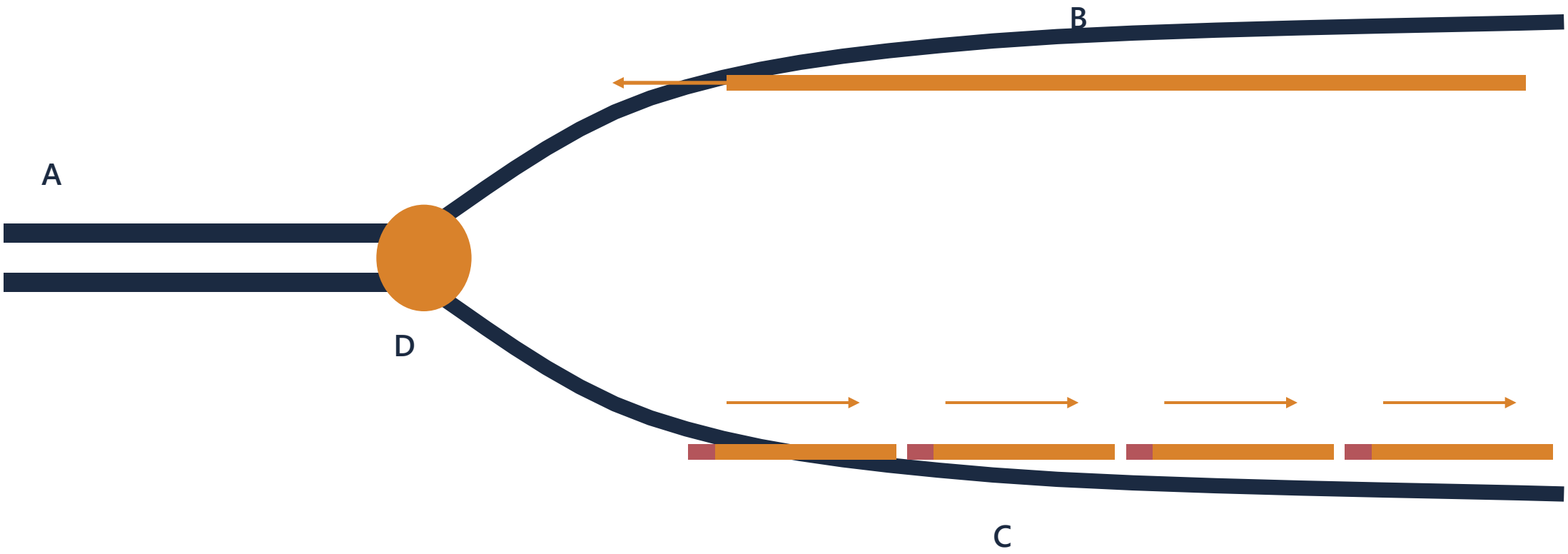
	Leading strand	Lagging strand
Synthesis	Continuous	Discontinuous
Direction	Towards the fork	Away from the fork
Primers	One	Many, one per fragment
Pieces	One long strand	Okazaki fragments

Remember

Both new strands grow 5′ to 3′. Only their direction relative to the fork differs.

Practice: label the fork

D1.1.6–7



Name A to D, and label the 5′ and 3′ ends of each new strand.

Check-in: class 1

D1.1.6–7

1. State the direction in which DNA polymerase builds new DNA.
2. Explain why it cannot build in the other direction.
3. Explain why the strands being antiparallel matters.
4. Distinguish between the leading and lagging strands.
5. Define Okazaki fragment.

How to use this

Answer without your notes, then check against slides 3–8.

The enzymes of replication

D1.1.8

Enzyme	Role (prokaryotes)
Helicase	Unwinds the helix; breaks hydrogen bonds between strands
DNA primase	Makes a short RNA primer: a 3′ end to build on
DNA polymerase III	Adds DNA nucleotides 5′ to 3′ by base pairing
DNA polymerase I	Removes RNA primers; replaces them with DNA
DNA ligase	Joins the backbones of neighbouring fragments

Lagging strand order

Primase, then polymerase III, then polymerase I, then ligase.

Making one Okazaki fragment

D1.1.8



1. Primase lays an RNA primer

2. Polymerase III extends it with DNA

3. Polymerase I swaps the primer for DNA

4. Ligase seals the gap

Drawn, not to scale. Red: RNA primer. Orange: DNA. Red dot: gap in the backbone.

Practice: what if an enzyme is missing?

D1.1.8

No primase

A mutant bacterium makes no primase.

No ligase

A mutant makes no DNA ligase.

No polymerase I

A mutant makes no DNA polymerase I.

For each one

Predict what would be found in the new DNA, and explain why.

Prokaryotes and eukaryotes

D1.1.8

	Prokaryotes	Eukaryotes
Origins	One per circular chromosome	Many per chromosome
Forks	Two	Many
Enzymes	Polymerases I and III, primase, ligase	More polymerases, same principles

Why many origins?

A human chromosome is far longer than a bacterial one. With one origin, copying would take weeks; many origins finish it in hours.

Working with data: speed of replication

D1.1.8

Approximate values, rounded for practice.

Quantity	Value
E. coli genome	4.6 million base pairs
Fork speed in E. coli	1000 bases per second
Human chromosome 1	250 million base pairs
Fork speed in humans	50 bases per second

1. E. coli has one origin and two forks. Estimate the time to copy its genome.
2. Estimate the time for chromosome 1 with only one origin.
3. Explain why humans need many origins.

Check-in: class 2

D1.1.8

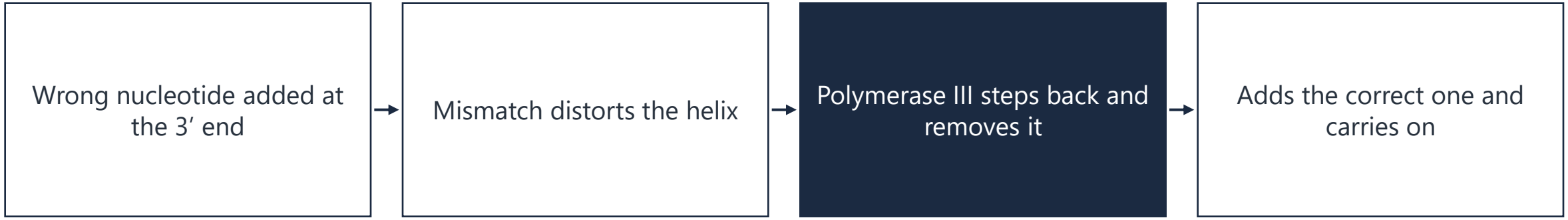
1. State the role of primase.
2. Explain why a primer is needed.
3. Distinguish between DNA polymerase I and III.
4. State the role of DNA ligase.
5. Explain why eukaryotes have many origins.

How to use this

Answer without your notes, then check against slides 10–14.

Proofreading by DNA polymerase III

D1.1.9



DNA polymerase III checks each base it adds. A mismatch is cut out (it works backwards, 3′ to 5′, to do this) and replaced.

Link to PCR

Taq polymerase does not proofread, so PCR copies can contain errors (D1.1.4).

Why proofreading matters

Stage	Errors per base
Base pairing alone	1 in 100 000
After proofreading	1 in 10 million
After later repair	1 in a billion

Scale

A human cell copies about 6 billion bases. Without proofreading, every division would add tens of thousands of mistakes.

Key idea

Accuracy keeps genes working; the rare errors that remain are new mutations.

Working with data: proofreading and errors

D1.1.9

Use the error rates on slide 17.

1. Calculate how many times proofreading improves accuracy.
2. The human genome has 3 billion base pairs. Estimate the errors per copy after all checks.
3. A bacterium lacks proofreading. Predict its mutation rate and one consequence.
4. Suggest why a low but non-zero error rate matters for evolution.

Practice: put it in order

D1.1.6–9

- (A) Ligase seals the backbone
- (B) Helicase separates the strands
- (C) Primase lays an RNA primer
- (D) Polymerase I replaces the primer
- (E) Polymerase III extends the primer and proofreads

Task

Put A to E in order for one Okazaki fragment, and say which steps also happen on the leading strand.

Check-in: class 3

1. Describe how DNA polymerase III proofreads.
2. Explain why proofreading matters.
3. Calculate the errors in a 4 million base genome at 1 error per 10 million bases.
4. Explain why PCR copies may contain errors.
5. Explain why some errors are useful for a species.

How to use this

Answer without your notes, then check against slides 16–19.

Common mistakes to avoid

Not this	But this
Polymerase III adds nucleotides to the 5′ end.	It adds them to the free 3′ end.
Primers are made of DNA.	Primers are RNA, made by primase.
Ligase makes the new strand.	Ligase only joins fragments.
The leading strand is made in fragments.	The lagging strand is made in fragments.

Summary: replication at HL

D1.1.6–9

	Key idea
D1.1.6	New DNA grows 5′ to 3′, added to the 3′ -OH
D1.1.7	Leading strand continuous; lagging strand in Okazaki fragments
D1.1.8	Helicase, primase, polymerase III, polymerase I, ligase
D1.1.9	Polymerase III proofreads: about 1 error in 10 million

Check that you can...

1. **Explain** the direction of DNA polymerase activity
2. **Distinguish** between the leading and lagging strands
3. **Outline** the roles of helicase, primase, polymerases I and III, and ligase
4. **Explain** proofreading by DNA polymerase III

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State the direction in which new DNA is synthesized.
2. Explain why the lagging strand is made as Okazaki fragments.
3. Outline the role of DNA polymerase I.

Image credits

DNA replication en.svg – LadyofHats Mariana Ruiz – Public domain

DNA chemical structure.svg – Madprime – CC BY-SA 3.0

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